

Probability that random points in a disk are in convex position.

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Abstract

In this paper we give a formula for the probability that n random points chosen under the uniform distribution in a disk are in convex position. While close, the formula is recursive and is totally explicit only for the first values of n .

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1 Introduction

All the random variables are assumed to be defined on a common probability space $(\Omega, \mathcal{A}, \mathbb{P})$. The expectation is denoted by $\mathbb{E}$. The plane will be sometimes viewed as $\mathbb{R}^2$ or as $\mathbb{C}$ and we will pass from the real notation (e.g. (x, y)) to the complex one ($\rho e^{i\theta}$) without any warning. For a set A in $\mathbb{R}^2$, $|A|$ denotes the Lebesgue measure of A . We denote by ∂B the border of a set B . For any $n \geq 1$, any z , notation $z[n]$ stands for the n tuple $(z_1, \dots, z_n)$ and $z\{n\}$ for the set $\{z_1, \dots, z_n\}$. For H be a compact convex domain in $\mathbb{R}^2$ with non empty interior, for any $n \geq 0$, $\mathbb{P}_H^n$ denotes the law of n i.i.d. points $z[n]$ taken under the uniform distribution over H . A n -tuple of points $x[n]$ of the plane is said to be in convex position if the x_i 's all belong to $\partial \text{ConvexHull}(x\{n\})$. Further we define

$$\text{CP}_{n,m} = \{x[n] : \#\{i : x_i \in \partial \text{ConvexHull}(x\{n\})\} = m\}$$

the set of n tuples $x[n]$ for which exactly m are on the border of $\partial \text{ConvexHull}(x\{n\})$. Hence $\text{CP}_n := \text{CP}_{n,n}$ is the set of n -tuples of points in convex position. Finally, we let

$$P_H^n = \mathbb{P}_H^n(z[n] \in \text{CP}_n), \quad (1)$$

$$P_H^{n,m} = \mathbb{P}_H^n(z[n] \in \text{CP}_{n,m}). \quad (2)$$

The aim of the paper is to establish a formula for P_D^n , the probability that n i.i.d. random points taken under the uniform distribution in a disk D are in convex position; we will also compute $P_D^{n,m}$ the probability that exactly m points among these n points are on $\partial \text{ConvexHull}(z\{n\})$. To compute P_D^n we need and obtain a result more general than the disk case only, result about for what we will call *bi-pointed segments* (BSEG). This will play somehow the role of the bi-pointed triangle (see (11)) as studied by Barany & al [3], central also in the approach of Buchta [6] (see (12)) of the computation of $P_T^{n,m}$ and $P_S^{n,m}$ (where T stands for triangle, and S for square).

For $\theta \in [0, 2\pi]$, $R > 0$, the *arc of circle* $AC(\theta, R)$ is defined by

$$AC(\theta, R) = \{Re^{i\nu}, \nu \in [-\theta/2, \theta/2]\}.$$

We denote by $SEG(\theta, R)$ the *segment* (SEG) corresponding to the convex hull of $AC(\theta, R)$ (see

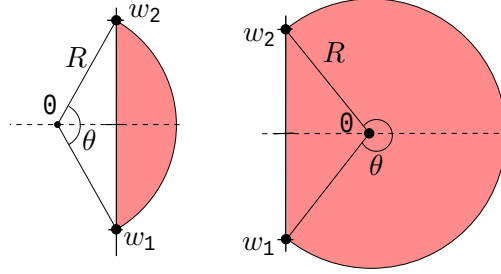


Figure 1: Representation of typical $SEG(\theta, R)$ for $0 < \theta < \pi$ and $\pi < \theta < 2\pi$

Fig. 1). Now consider $w_1(\theta, R) = Re^{-i\theta/2}$ and $w_2(\theta, R) = Re^{i\theta/2}$ the two *extremities* of the *special border* $[w_1(\theta, R), w_2(\theta, R)]$ of $SEG(\theta, R)$. Let $z_1, \dots, z_n$ be i.i.d. and uniform in $SEG(\theta, R)$. Set

$$Z[n, \theta, R] = \{w_1(\theta, R), w_2(\theta, R), z_1, \dots, z_n\},$$

and define the crucial *bi-pointed segment case* (BSEG) function

$$B_{n,m}(\theta) := \mathbb{P}(Z[n, \theta, R] \in \text{CP}_{n+2,m+2}), \quad \theta \in (0, 2\pi), 1 \leq m \leq n. \quad (3)$$

The value of R has no importance since there exists a dilatation sending $SEG(\theta, R)$ on $SEG(\theta, R')$, and dilatations conserve convex bodies and uniform distribution. But it will be useful to have the two parameters (θ, R) for subsequent computations. Again, we write B_n instead of $B_{n,n}$ and below L_n instead of $L_{n,n}$. Clearly, for any $\theta \in (0, 2\pi)$, $B_0(\theta) = B_1(\theta) = 1$. Now for any $n \geq 0$, $\theta \in (0, 2\pi)$ define

$$L_{n,m}(\theta) = \frac{B_{n,m}(\theta)(\theta - \sin(\theta))^n \sin(\theta/2)}{n!} \quad (4)$$

Hence

$$L_0(\theta) = \sin(\theta/2), \quad L_1(\theta) = \sin(\theta/2)(\theta - \sin(\theta)). \quad (5)$$

Notice that 0 (corresponding to the flat case) as well as 2π (the circle case) are excluded from definitions (3) and (4). The main contribution of this paper is the following theorem which allows to compute $P_D^{n,m}$.

Theorem 1 (i) For any $n \geq 1$,

$$P_D^n = \lim_{\substack{t \rightarrow 2\pi \\ t < 2\pi}} B_{n-1}(t).$$

(i') For any $n \geq 2$,

$$P_D^n = \frac{(n-2)!}{2^{n-2}\pi^{n-1}} \int_0^{2\pi} \sum_{k=0}^{n-2} L_k(\phi) L_{n-2-k}(2\pi - \phi) d\phi$$

(ii) For any $\theta \in (0, 2\pi)$, any $n \geq 1$,

$$\frac{L_n(\theta)}{2} = \int_0^\theta \frac{\sin(\theta/2)^{2n+1}}{\sin(\phi/2)^{2n+1}} \int_0^\phi \sum_{k=0}^{n-1} L_k(\eta) L_{n-1-k}(\phi - \eta) d\eta d\phi. \quad (6)$$

Analogous results can be obtained for $P_D^{n,m}$:

(iii) For any $\theta \in (0, 2\pi)$, for any $k, l \geq k+1$, $L_{k,l}(\theta) = 0$. For any $\theta \in (0, 2\pi)$, any $n \geq 1$ and $1 \leq m \leq n$

$$\begin{aligned} \frac{L_{n,m}(\theta)}{2} &= \int_0^\theta \int_0^\phi \frac{\sin(\theta/2)^{2n+1}}{\sin(\phi/2)^{2n+1}} \sum_{\substack{n_1+n_2+n_3=n-1 \\ m_1+m_2=m-1}} \frac{(\sin(\eta) + \sin(\phi - \eta) - \sin(\phi))^{n_3}}{n_3!} \\ &\quad \times L_{n_1,m_1}(\eta) L_{n_2,m_2}(\phi - \eta) d\eta d\phi \end{aligned}$$

An alternative form can be given using

$$\sin(\eta) + \sin(\phi - \eta) - \sin(\phi) = 4 \sin\left(\frac{\phi - \eta}{2}\right) \sin(\phi/2) \sin(\eta/2).$$

(iii') For any $n \geq 2$, $1 \leq m \leq n$

$$P_D^{n,m} = \frac{(n-2)!}{2^{n-2}\pi^{n-1}} \int_0^{2\pi} \sum_{\substack{n_1+n_2=n-1 \\ m_1+m_2=m-1}} L_{n_1,m_1}(\phi) L_{n_2,m_2}(2\pi - \phi) d\phi.$$

(iv) For any $n \geq 1$,

$$P_D^{n,m} = \lim_{\substack{t \rightarrow 2\pi \\ t < 2\pi}} B_{n-1,m-1}(t).$$

From (ii), one can compute successively the $L_j(\theta)$, and by (4), this allows one to compute the $B_n(\theta)$. By (i) it suffices then to take the limit when $\theta \rightarrow 2\pi^-$ (limit from below).

Despite important efforts we were not able to find a simpler formula for B_n than that presented in the Theorem. Nevertheless, explicit computation can be done but close formula for the first L_j given below shows a rapid growth of the complexity of the formula's (L_{10} would need one page to be written down). The effective computation of the first L_n is complex and very few can be computed by hand. In particular the singularity apparent in (6) is difficult to handle since the terms in the sum needs to be combined to compensate the singularity.

In Section 3 we present an algorithm allowing one to compute the first terms of the sequence (using a computer, or years of time of an efficient human brain). With this algorithm I computed

the 11 rst values of L_n which allows the computation ($P_D^n, 1 \leq n \leq 12$). L_0 and L_1 have been given in (5); the next ones are

$$\begin{aligned}
L_2(\theta) &= \frac{1}{6} \sin\left(\frac{\theta}{2}\right)(3\theta^2 + \sin(\theta)^2 - 16 \sin\left(\frac{\theta}{2}\right)^2) \\
L_3(\theta) &= \frac{1}{54} \sin\left(\frac{\theta}{2}\right) \left(2 \sin\left(\frac{\theta}{2}\right)^4 \sin(\theta) + 9\theta^3 + 27 \sin\left(\frac{\theta}{2}\right)^2 \theta + 7 \sin\left(\frac{\theta}{2}\right)^2 \sin(\theta) + 105(\sin(\theta) - \theta) \right) \\
L_4(\theta) &= \frac{\sin\left(\frac{\theta}{2}\right)}{12960} \left(160 \sin\left(\frac{\theta}{2}\right)^6 + 48 \sin\left(\frac{\theta}{2}\right)^8 + 60 \sin\left(\frac{\theta}{2}\right)^4 + 540 \theta^4 - 13725 \theta^2 - 7200 \sin(\theta) \theta \right. \\
&\quad \left. + 83700 \sin\left(\frac{\theta}{2}\right)^2 \right) \\
L_5(\theta) &= \frac{\sin\left(\frac{\theta}{2}\right)}{1296000} \left(40500 \sin(\theta) \theta^2 - 584 \sin\left(\frac{\theta}{2}\right)^4 \sin(\theta) + 12000 \sin\left(\frac{\theta}{2}\right)^4 \theta \right. \\
&\quad \left. - 272 \sin\left(\frac{\theta}{2}\right)^6 \sin(\theta) - 549000 \theta^3 - 2745000 \sin\left(\frac{\theta}{2}\right)^2 \theta \right. \\
&\quad \left. + 44270 \sin\left(\frac{\theta}{2}\right)^2 \sin(\theta) + 7102095(\theta - \sin(\theta)) - 64 \sin\left(\frac{\theta}{2}\right)^8 \sin(\theta) + 10800 \theta^5 \right)
\end{aligned}$$

The next formula are too large to be written here. We can compute also $L_{m,n}(\theta)$ for small values of m, n . For any $n \geq 2$, $\sum_{k=1}^n B_{n,k}(\theta) = 1$. Since $B_{2,2} = B_2$ is know, so do $B_{2,1}$. The next ones are

$$\begin{aligned}
L_{3,1}(\theta) &= -\frac{1}{6} \sin\left(\frac{\theta}{2}\right)^3 \left(48 \sin\left(\frac{\theta}{2}\right)^5 \cos\left(\frac{\theta}{2}\right) - 48 \sin\left(\frac{\theta}{2}\right)^6 - 12 \sin\left(\frac{\theta}{2}\right)^3 \cos\left(\frac{\theta}{2}\right) + 8 \sin\left(\frac{\theta}{2}\right)^4 \right. \\
&\quad \left. + 12 \sin\left(\frac{\theta}{2}\right)^2 \theta - 10 \cos\left(\frac{\theta}{2}\right) \sin\left(\frac{\theta}{2}\right) + 40 \sin\left(\frac{\theta}{2}\right)^2 - 15 \theta \right) \\
L_{4,1}(\theta) &= -\frac{1}{45} \sin\left(\frac{\theta}{2}\right)^4 \left(48 \sin\left(\frac{\theta}{2}\right)^4 \cos\left(\frac{\theta}{2}\right) \theta - 48 \sin\left(\frac{\theta}{2}\right)^5 \theta - 12 \sin\left(\frac{\theta}{2}\right)^5 - 12 \cos\left(\frac{\theta}{2}\right) \sin\left(\frac{\theta}{2}\right)^2 \theta \right. \\
&\quad \left. - 48 \sin\left(\frac{\theta}{2}\right)^3 \theta + 190 \sin\left(\frac{\theta}{2}\right)^3 + 9 \cos\left(\frac{\theta}{2}\right) \theta + 96 \sin\left(\frac{\theta}{2}\right) \theta - 210 \sin\left(\frac{\theta}{2}\right) \right)
\end{aligned}$$

the next ones are too large again, to be written here. I am able to compute $L_{n,m}$ for $n \leq 5$ (which provides the values of $P_D^{n,m}$ for $n \leq 7$).

Using these formulae, one finds the following explicit cute values for P_D^n

n	4	5	6	7	8
$1 - P_D^n$	$\frac{35}{12\pi^2}$	$\frac{305}{48\pi^2}$	$\frac{146400\pi^2 - 473473}{11520\pi^4}$	$\frac{512400\pi^2 - 2900611}{23040\pi^4}$	$\frac{62664108221 + 1721664000\pi^4 - 18670881600\pi^2}{48384000\pi^6}$

(7)

By Theorem 1, we can also compute $P_D^{4,3} = \frac{35}{12\pi^2}$ (or by $P_D^{4,4} + P_D^{4,3} = 1$), $P_D^{5,5}$ is given in the array, $P_D^{5,4} = \frac{65}{12\pi^2}$, $P_D^{5,3} = \frac{15}{16\pi^2}$, $P_D^{6,5} = \frac{3120\pi^2 - 17017}{288\pi^4}$, $P_D^{6,4} = \frac{7200\pi^2 + 57057}{3840\pi^4}$, $P_D^{6,3} = \frac{1001}{320\pi^4}$.

Explicit results for bi-pointed half disk ($B_1(\theta) = 1$ for any $\theta \in (0, 2\pi)$):

n	2	3	4	5	6
$1 - B_n(\pi)$	$\frac{16}{3\pi^2}$	$\frac{26}{3\pi^2}$	$\frac{-83968 + 13725\pi^2}{540\pi^4}$	$\frac{-97091 + 12200\pi^2}{240\pi^4}$	$\frac{30749622272 - 4885982325\pi^2 + 201757500\pi^4}{2268000\pi^6}$

(8)

The only simple formula which appears is the following:

$$\lim_{t \rightarrow 0} B_n(t) = \frac{12^n}{(n+1)(2n+1)!}.$$

which holds for all $1 \leq n \leq 11$. It corresponds to the limit for BSEG with an angle going to 0.

Apart the results exposed in Theorem 1, the only explicit results in the literature concerns triangles and parallelogram (we here discuss only results known for any n , in 2D). Valtr [13] (1995) has obtained that if S is a square (or a (non at) parallelogram) then, for $n \geq 1$,

$$P_S^n = \left(\frac{\binom{2n-2}{n-1}}{n!} \right)^2, \quad (9)$$

and in a second paper, [14] (1996) he proved that if T is a (non at) triangle then, for $n \geq 1$,

$$P_T^n = \frac{2^n(3n-3)!}{(n-1)!^3(2n)!}. \quad (10)$$

Buchta [6] goes further and gives an expression for $P_S^{n,m}$ and $P_P^{n,m}$ as a finite sum of explicit terms.

For the bi-pointed triangle, Barany, Rote, Steiger, Zhang [3] (2000) have shown the following. Let $T = (A, B, C)$ be a (non at) triangle, and let $(z_1, \dots, z_n)$ be $\mathbb{P}_T^n$ distributed, and let $\overline{z[n]} = (A, B, z_1, \dots, z_n)$ be the $n+2$ tuple obtained by adding A, B to $z[n]$. For any $n \geq 0$,

$$\mathbb{P}_T^n(\overline{z[n]} \in \text{CP}_{n+2}) = \frac{2^n}{n!(n+1)!}. \quad (11)$$

These results are at the origin of numerous works concerning limit shape for convex bodies in a domain ([3], Barany [1]) and for the evaluation of the probability that n points chosen in a convex domain H are in convex position (see Barany [1]).

Buchta (2007) [5] goes forward and prove the following result : For any $n \geq 1$, any $1 \leq m \leq n$,

$$\mathbb{P}_T^n(\overline{z[n]} \in \text{CP}_{n+2,m+2}) = \sum_{C \in \text{Comp}(n,m)} 2^m \prod_{i=1}^m \frac{C_i}{SC_i(1+SC_i)} \quad (12)$$

where $SC_i = C_1 + \dots + C_i$ and $\text{Comp}(n, m)$ is the set of compositions of n in m non empty parts (Examples : $\text{Comp}(2, 3) = \emptyset$, $\text{Comp}(4, 2) = \{(1, 3), (3, 1), (2, 2)\}$).

Additional references The literature concerning the question of the number of points on the convex hull for i.i.d. random points taken in a convex domain is huge. I won't make a survey here but rather sends the interested reader to Reitzner [11], Hug [9] and to the various paper cited in the present paper I will focus on what concerns the disk.

Blaschke (1917) [4] proves that for the 4 points problem (the so-called *problem of Sylvester*), we have for any convex K ,

$$P_T^4 \leq P_K^4 \leq P_D^4.$$

Barany (2000) [1] have shown that

$$\lim_{n \rightarrow +\infty} n^2 (P_K^n)^{1/n} = e^2 A^3(K)/4 \quad (13)$$

where $A^3(K)$ is the supremum of the *average perimeter* of all convex sets $S \subset K$. For the disk one gets

$$\log(P_K^n) = -2n \log n + n \log(2\pi^2 e^2) - 2\varepsilon_0(3\pi^4 n)^{1/5} + \dots \quad (14)$$

where the last term, not really proved in the mathematical sense, has been obtained by Hilhorst & al. [8]. Central limit theorems exists also for the number of points on $\partial \text{ConvexHull}(\mathbf{x}\{n\})$ under $\mathbb{P}_D^n$ (and for more general domain, under the uniform or Poisson distribution), see Groeneboom [7], Pardon [10], Barany and Reitzner [2].

2 Proof of Theorem 1

2.1 Proof of ()

All along this section $n \geq 1$ is fixed. Take a closed disk $\bar{B} = \bar{B}((0,0), R_c)$, with center $(0,0)$ and radius $R_c = 1/\sqrt{\pi}$, that is with area 1, and pick n i.i.d. uniform points $U_1, \dots, U_n$ in $\bar{B}$. Now consider the smallest disk $\bar{B}((0,0), R_n)$ that contains all the U_i 's. Clearly

$$R_n = \inf\{r : \#(\bar{B}(0, r) \cap \{U_1, \dots, U_n\}) = n\}.$$

Proposition 2 *Conditionally on $R_n = r$, there is a.s. exactly one index $J \in \llbracket 1, n \rrbracket$ such that U_J belongs to the circle $B((0,0), r)$. Conditionally on $\{J = j, R_n = r\}$, U_j and $(U_1, \dots, U_{j-1}, U_{j+1}, \dots, U_n)$ are independent, U_j has the uniform law on the circle $\partial B((0,0), r)$, and $U_1, \dots, U_{j-1}, U_{j+1}, \dots, U_n$ are uniform in $\bar{B}((0,0), r)$.*

Proof. A.s. the points $U_1, \dots, U_n$ are not on the same circle with center $(0,0)$, and by symmetry conditionally on $R_n = r$ and $J = j$, U_j is uniform on $B((0,0), r)$. Now, conditionally on $R_n = r$ and $J = j$, each variable U_ℓ (for $\ell \neq j$) are just conditioned to satisfy $\|U_\ell\|_2 \leq r$, and this conditioning conserves the uniform distribution. $\square$

Proof of Theorem 1(i). Theorem 1(i) is { or should be { intuitively obvious, taking into account Proposition 2. This Proposition says that the two following models (a) and (b):

{ (a) n points i.i.d. uniform in a disk,

{ (b) one point uniform on the circle and, independently, $n - 1$ i.i.d. uniform inside the disk are equivalent with respect to the probability to be in convex position.

Now if we come back to the BSEG considerations, when $\theta \rightarrow 2\pi$, the points $w_1(R_c, \theta)$ and $w_2(R_c, \theta)$ become closer and closer, and the line passing by these points lets all the other points in one of the half plane it defines. It is intuitively clear that replacing $w_1(R_c, \theta)$ and $w_2(R_c, \theta)$ by a single point close to them (for example, at position $(-R_c, 0)$) will not dramatically change the model nor the probability to be in convex position. This is the essence of Theorem 1(i).

For sake of completeness, let us give a formal proof. Take $R > 0$ and consider the two sets $S(\varepsilon) = \text{SEG}(2\pi - \varepsilon, R)$ and $S = \text{SEG}(2\pi, R) = \overline{B}((0, 0), R)$. These two sets are close for the Hausdorff topology when ε is small. We always have $S(\varepsilon) \subset S$, and $|S \setminus S(\varepsilon)|$ goes to 0. This property implies that if we $\forall \varepsilon' > 0$, for ε small enough, for $z_1, \dots, z_n$ chosen uniformly and independently under $\mathbb{P}_S$,

$$\mathbb{P}(\{z_1, \dots, z_n\} \subset S(\varepsilon)) \geq 1 - \varepsilon'. \quad (15)$$

Conditionally on the event $\varepsilon := \{\{z_1, \dots, z_n\} \subset S(\varepsilon)\}$, the z_i 's are i.i.d. uniform in $S(\varepsilon)$. Let $w_1(\varepsilon) = w_1(R_c, 2\pi - \varepsilon)$, $w_2(\varepsilon) = w_2(R_c, 2\pi - \varepsilon)$, $w = -R$.

We want to show that $\mathbb{P}((z_1, \dots, z_n, w_1^\varepsilon, w_2^\varepsilon) \in \text{CP}_{n+2} \mid \varepsilon) \rightarrow \mathbb{P}((z_1, \dots, z_n, w) \in \text{CP}_{n+1})$. Consider the following sets (subsets of S^n):

$$\begin{aligned} E_1(\varepsilon) &:= \{(t_1, \dots, t_n) \in S(\varepsilon) : (t_1, \dots, t_n, w_1(\varepsilon), w_2(\varepsilon)) \in \text{CP}_{n+2}\} \\ E_2 &:= \{(t_1, \dots, t_n) \in S : (t_1, \dots, t_n, w) \in \text{CP}_{n+1}\}. \end{aligned}$$

It suffices to prove that $|E_1(\varepsilon)| \xrightarrow{\varepsilon \rightarrow 0} |E_2|$. First $E_1(\varepsilon) \subset E_2$ since if $(t_1, \dots, t_n, w_1(\varepsilon), w_2(\varepsilon))$ belongs to CP_{n+2} and since the segments $[w_1(\varepsilon), w]$ and $[w_2(\varepsilon), w]$ are chords, then $(z_1, \dots, z_n, w_1(\varepsilon), w_2(\varepsilon), w)$ is in CP_{n+3} from what we deduce that E_2 is in CP_{n+1} .

To end the proof take $(t_1, \dots, t_n) \in E_2$. We show that when ε is small enough, it is in $E_1(\varepsilon)$. More precisely, we will see that it is not the case only if the t_i belongs to a null set (for Lebesgue measure). We assume that $n \geq 2$ since for $n = 1$ the result is clear.

First, for $\varepsilon > 0$ small enough, if the t_i 's are different and different to $-R$, all the t_i belongs to $S(\varepsilon)$. Since $(t_1, \dots, t_n, w) \in \text{CP}_{n+1}$ draw the convex polygon p passing by these points, and relabel the t_i 's as $t_1^*, \dots, t_n^*$ clockwise around p so that the neighbours of w are t_1^* and t_n^* . Again, up to null set, the angles (w, t_1^*, t_2^*) and (t_{n-1}^*, t_n^*, w) are not 0, and it appears clearly that for ε small enough, $(t_1, \dots, t_n, w_1(\varepsilon), w_2(\varepsilon)) \in \text{CP}_{n+2}$. We then have $E_2 = \cup_\varepsilon E_1(\varepsilon)$ and the $E_1(x) \cup E_1(x')$ if $x' < x$, so $|E_1(\varepsilon)| \rightarrow |E_2|$ when ε goes to 0. $\square$.

2.2 Proof of ()

For any $\theta \in [0, 2\pi]$, $R > 0$,

$$|\text{SEG}(\theta, R)| := \frac{R^2}{2} (\theta - \sin(\theta)) \quad (16)$$

and then for

$$R_\theta = \sqrt{\frac{2}{\theta - \sin(\theta)}}, \quad (17)$$

the area $|\text{SEG}(\theta, R_\theta)| = 1$. Denote more simply by SEG_θ the segment $\text{SEG}(\theta, R_\theta)$ with unit area. The size L_θ of the special border $[w_1(\theta, R_\theta), w_2(\theta, R_\theta)]$ for this segment is

$$L_\theta = 2R_\theta \sin(\theta/2). \quad (18)$$

In this section we fix $\theta \in (0, 2\pi)$ and search to express $B_n(\theta)$ with some combinations of $B_j(\nu)$, for $\nu < \theta$ and $j < n$. To get the decomposition we will "push the arc of circle" $AC(\theta, R)$ inside $SEG(\theta, R_\theta)$ till it touches one of the z_i 's doing something similar to the Buchta's method (for the computation of P_S^n and P_T^n). Here it is a bit more complex: we need the arc of circle to stay an arc of circle during the operation in order to get a nice decomposition, and also we somehow need to keep the bi-pointed elements. The arc angle and radius will change during the operation. This will lead to a quadratic formula for B_n . Almost of quantities appearing in this section should be

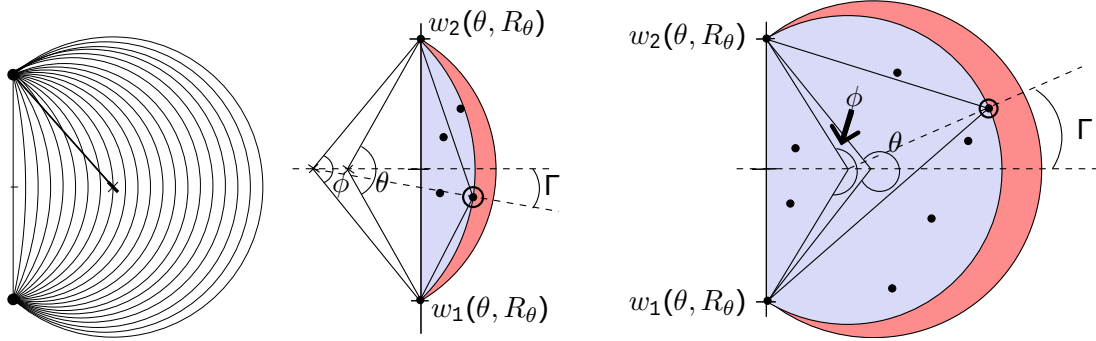


Figure 2: Representation of the family $\mathcal{F}_\theta$. The angle $\phi < \theta$ and $SEG[\phi] \leq SEG[\theta]$. The angles are taken at the center of the circle that defines the segments.

indexed by θ . In order to avoid heavy notation we won't do this. Draw SEG_θ in the plane. We consider the family of segments

$$\mathcal{F}_\theta := (SEG[\phi], 0 \leq \phi \leq \theta)$$

having as special border the special border of SEG_θ , that is $[w_1(\theta, R_\theta), w_2(\theta, R_\theta)]$, and lying at its right, such that the angle of $SEG[\phi]$ is ϕ (see Fig. 2).

When ϕ goes from θ to 0, the center $O[\phi]$ of (the circle which defines) $SEG[\phi]$ moves on the x -axis from $O[\theta] = 0$ to $(-\infty, 0)$. Comparing the distance from $O[\phi]$ to the special border, we can compute the coordinate of $O[\phi]$:

$$O[\phi] = \frac{L_\theta}{2} \left(\cot\left(\frac{\theta}{2}\right) - \cot\left(\frac{\phi}{2}\right) \right) \quad (19)$$

and the radius of $SEG[\phi]$,

$$R[\phi] = R_\theta \frac{\sin(\theta/2)}{\sin(\phi/2)}. \quad (20)$$

Since the special border of all the $SEG[\phi]$ is the same one sees that if $\phi < \phi'$ then $SEG[\phi] \subset SEG[\phi']$. When ϕ goes to 0, $SEG[\phi]$ goes to $[w_1(\theta, R_\theta), w_2(\theta, R_\theta)]$ (for the Hausdorff topology). One also sees that $SEG[\theta] = SEG_\theta$, and for $\phi < \theta$, by (16) and (17),

$$|SEG[\phi]| = \left(\frac{\sin(\theta/2)}{\sin(\phi/2)} \right)^2 \frac{\phi - \sin(\phi)}{\theta - \sin(\theta)} \quad (21)$$

and then the other segments of the family $\mathcal{F}_\theta$ have area smaller than 1 (see Fig. 2).

Again θ is fixed. Let $z_1, \dots, z_n$ be $n \geq 1$ i.i.d. uniform random points in SEG_θ . Denote by

$$= \min \{ \phi : \#(\{z_1, \dots, z_n\} \cap \text{SEG}[\phi]) = n \},$$

and let J the (a.s. unique) index of the variable z_j on $\partial \text{SEG}[\phi]$. Finally let γ be the (signed) angle $((+\infty, 0), O[\gamma], z_J)$ formed by the x -axis and the line $(O[\gamma], z_J)$ (see Fig. 2). We have

Proposition 3 *The distribution of (ϕ, γ) admits the following density $f_{(\phi, \gamma)}$ with respect to the Lebesgue measure*

$$f_{(\phi, \gamma)}(\phi, \gamma) = n \frac{\sin(\theta/2)^{2n}}{(\theta - \sin(\theta))^n} \frac{(\phi - \sin(\phi))^{n-1}}{\sin(\phi/2)^{2n+1}} (\cos(\gamma) - \cos(\phi/2)) \mathbf{1}_{0 \leq \phi \leq \theta} \mathbf{1}_{|\gamma| \leq \phi/2}.$$

Proof. First, the density of $z_J = (x, y)$ with respect to the Lebesgue measure on $|\text{SEG}_\theta|$ is $ndxdy|\text{SEG}_{x,y}|^{n-1}$ where $|\text{SEG}_{x,y}|^{n-1}$ is the area of the unique element of the family $\mathcal{F}_\theta$ whose border contains (x, y) . We then just have to make a change of variables in this formula !

We search the unique pair (ϕ, γ) such that

$$x + iy = R[\phi]e^{i\gamma} + O[\phi].$$

Since by (21) and (19) everything is explicit, we can compute the Jacobian

$$\left| \det \begin{pmatrix} \frac{\partial x}{\partial \phi} & \frac{\partial x}{\partial \gamma} \\ \frac{\partial y}{\partial \phi} & \frac{\partial y}{\partial \gamma} \end{pmatrix} \right| = \frac{\sin(\theta/2)^2 (\cos(\gamma) - \cos(\phi/2))}{\sin(\phi/2)^3 (\theta - \sin(\theta))}.$$

From what we deduce the wanted formula, using (21). $\square$.

Now, it remains to end the decomposition of our problem. Conditionally on $(\phi, \gamma, j) = (\phi, \gamma, j)$, the points $z_1, \dots, z_{j-1}, z_{j+1}, \dots, z_n$ are i.i.d. uniform in $\text{SEG}[\phi]$.

The triangle $T := (w_1(\theta, R_\theta), w_2(\theta, R_\theta), z_j)$ is inscribed in $\text{SEG}[\phi]$ and $\text{SEG}[\phi] \setminus T$ produces two segments S_1 and S_2 . Since we may rescale $\text{SEG}[\phi]$ to be SEG_ϕ (to get area 1), the question now is that of the area of the two rescaled segments. After rescaling, S_1 and S_2 appear to be $\text{SEG}[\phi/2 + \gamma, R_\phi]$ and $\text{SEG}[\phi/2 + \gamma, R_\phi]$ since these lards have the right angles. Using (16)

$$|\text{SEG}[\alpha, R_\phi]| = \frac{\alpha - \sin(\alpha)}{\phi - \sin(\phi)}. \quad (22)$$

We keep temporarily notation S_1 and S_2 instead of $\text{SEG}[\phi/2 + \gamma, R_\phi]$ and $\text{SEG}[\phi/2 + \gamma, R_\phi]$ for short. The following Proposition is a simple consequence of the fact that the uniform distribution is conserved by conditioning. It is the "combinatorial decomposition" of the computation of $B_n(\theta)$, illustrated on Fig. 3.

Proposition 4 (i) *Conditionally on $(\phi, \gamma, j) = (\phi, \gamma, j)$, the respective number (N_1, N_2, N_3) of points of $z\{n\} \setminus \{z_j\}$ in S_1, S_2 and $\text{SEG}_\phi - (S_1 \cup S_2)$ is*

$$\text{Multinomial}(n-1, |S_1|, |S_2|, 1 - |S_1| - |S_2|).$$

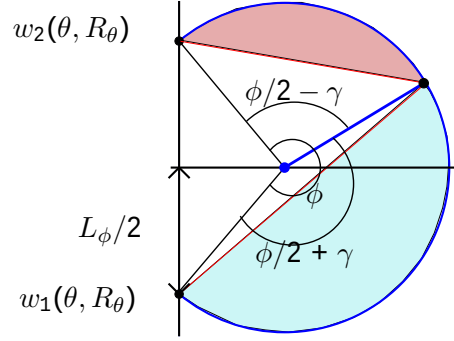


Figure 3: Decomposition of the computation of $B_n(\theta)$, and definition of the two sub-segments appearing in the decomposition.

(ii) Conditionally on $(\phi, \gamma, j) = (\phi, \gamma, j)$ and $(N_1, N_2, N_3) = (k_1, k_2, k_3)$ the points $z_1, \dots, z_n$ are in convex position with probability $1_{k_3=0, k_1+k_2=n-1} B_{k_1}(\phi/2 + \gamma) B_{k_2}(\phi/2 - \gamma)$.

Putting everything together we have obtained

$$B_n(\theta) = \int_0^\theta \int_{-\frac{\phi}{2}}^{\frac{\phi}{2}} f(\phi, \gamma) \sum_{k=0}^{n-1} \binom{n-1}{k} |S_1|^k |S_2|^{n-1-k} B_k(\phi/2 + \gamma) B_{n-1-k}(\phi/2 - \gamma) d\gamma d\phi$$

Set $\eta = \phi/2 + \gamma$, $d\eta = d\gamma$, η goes from 0 to ϕ (and $\phi/2 - \gamma = \phi - \eta$), giving

$$B_n(\theta) = \int_0^\theta \int_0^\phi f(\phi, \eta - \phi/2) \sum_{k=0}^{n-1} \binom{n-1}{k} \quad (23)$$

$$\times |\text{SEG}[\eta, R_\phi]|^k |\text{SEG}[\phi - \eta, R_\phi]|^{n-1-k} B_k(\eta) B_{n-1-k}(\phi - \eta) d\eta d\phi \quad (24)$$

from which we get

$$B_n(\theta) = \int_0^\theta \int_0^\phi n \frac{\sin(\theta/2)^{2n}}{(\theta - \sin(\theta))^n} \frac{\cos(\eta - \phi/2) - \cos(\phi/2)}{\sin(\phi/2)^{2n+1}} \sum_{k=0}^{n-1} \binom{n-1}{k} \quad (25)$$

$$\times (\eta - \sin(\eta))^k B_k(\eta) ((\phi - \eta) - \sin(\phi - \eta))^{n-1-k} B_{n-1-k}(\phi - \eta) d\eta d\phi. \quad (26)$$

Now, $\cos(\eta - \phi/2) - \cos(\phi/2) = 2 \sin(\eta/2) \sin((\phi - \eta)/2)$. Finally setting $L_n(\theta)$ as done in (4), we obtain Formula 1(ii).

2.3 Proof (')

Recall Proposition 2. To compute P_n^D we can work under the model where $n - 1$ points $z_1, \dots, z_{n-1}$ are picked independently and uniformly inside the disk $B((0, 0), R_c)$ (with $R_c = \pi^{-1/2}$) and one point on the border. We place this last point at position $-R_c$ which is allowed since rotation keeps convex bodies and the uniform distribution.

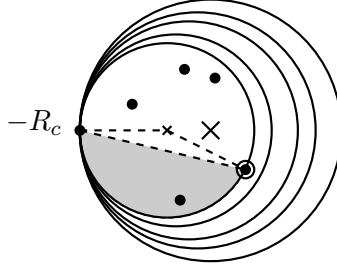


Figure 4: Decomposition of the computation of P_n^D . The big cross is the center of the initial circle, the small one, the center of the smallest circle containing all the points.

Now take a family of circles $\mathcal{G} = \{B[r], 0 \leq r \leq R_c\}$ such that $B[r]$ as radius r , its center at position $-R_c + r$, implying that $-R_c$ belongs to all these circles (see Fig. 4).

If $r' < r$, $B[r'] \subset B[r]$. Let r^* be the largest circle such that exists $1 \leq k \leq n-1$, $z_k \in \partial B[r^*]$. Denote then by ϕ the angle such that $z_k = (-R_c + r) + re^{i(-\pi+\phi)}$. If we denote by (X, Y) the (Euclidean) position of z_k , the density of the distribution of (x, y) is

$$(n-1)1_{(x,y) \in B((0,0), R_c)} |B[r]|^{n-2} dx dy$$

where $B[r]$ is the unique circle in the family $\mathcal{G}$ which passes by (x, y) . We can then compute the Jacobian and find the distribution of (r, ϕ) to be with density $1_{0 \leq r \leq R_c, 0 \leq \phi \leq 2\pi} r(1 - \cos(\phi))(\pi r^2)^{n-2} dr d\phi$. Once z_k is given, we can once again normalise the problem, and come back on a circle of area R_c . We then get, using $1 + \cos(\phi) = 2 \sin^2(\phi/2)$

$$\begin{aligned} P_D^n &= (n-1) \int_0^{R_c} \int_0^{2\pi} \sum_{k=0}^{n-2} \binom{n-2}{k} 2 \sin^2(\phi/2) r (\pi r^2)^{n-2} \\ &\quad \times B_k(\phi) B_{n-2-k}(2\pi - \phi) |\text{SEG}(\phi, R_c)|^k |\text{SEG}(2\pi - \phi, R_c)|^{n-2-k} d\phi dr \end{aligned}$$

The integration with respect to dr gives

$$\begin{aligned} P_D^n &= \frac{1}{\pi} \int_0^{2\pi} \sum_{k=0}^{n-2} \binom{n-2}{k} \sin^2(\phi/2) B_k(\phi) B_{n-2-k}(2\pi - \phi) \\ &\quad \times \left(\frac{\phi - \sin(\phi)}{2\pi} \right)^k \left(\frac{2\pi - \phi + \sin(\phi)}{2\pi} \right)^{n-2-k} d\phi \end{aligned}$$

since once ϕ is known, the convexity follows that on the pair of bi-pointed segments with angles ϕ and $2\pi - \phi$, and the number of elements in these segments is binomial $(n-2, |\text{SEG}(\phi, R_c)|)$.

2.4 Proof of ()

The proof is the same as that of (ii) except that in Proposition 4 we need to follow the number of points falling in the triangle. We then get

$$\begin{aligned}
B_{n,m}(\theta) &= \int_0^\theta \int_0^\phi f(\cdot, \cdot)(\phi, \eta - \phi/2) \sum_{\substack{n_1+n_2+n_3=n-1 \\ m_1+m_2=m-1}} \binom{n-1}{n_1, n_2, n_3} \\
&\times |\text{SEG}[\eta, R_\phi]|^{n_1} |\text{SEG}[\phi - \eta, R_\phi]|^{n_2} (1 - |\text{SEG}[\eta, R_\phi]| - |\text{SEG}[\phi - \eta, R_\phi]|)^{n_3} \\
&\times B_{n_1, m_1}(\eta) B_{n_2, m_2}(\phi - \eta) d\eta d\phi
\end{aligned}$$

Using the notation introduced in (4) we get (iii) .

2.5 Proof of (')

Copy the arguments in Section 2.3. In the same way, for $n \geq 2$, $1 \leq m \leq n$

$$\begin{aligned}
P_D^{n,m} &= \frac{1}{\pi} \int_0^{2\pi} \sum_{k=0}^{n-2} \sum_{1 \leq m_1 \leq n-2} \binom{n-2}{k} \sin^2(\phi/2) \\
&\times B_{k, m_1}(\phi) B_{n-2-k, m-m_1-2}(2\pi - \phi) \left(\frac{\phi - \sin(\phi)}{2\pi} \right)^k \left(\frac{2\pi - \phi + \sin(\phi)}{2\pi} \right)^{n-2-k} d\phi
\end{aligned}$$

with the condition that $B_{k, k+l} = 0$. (iii') follows.

2.6 Proof of ()

The same proof of (i) does the job.

3 Effective computation of B_n

We explain in this part how to make effective computations. Since $B_1(\theta) = B_0(\theta) = 1$, $L_0(\theta)$ and $L_1(\theta)$ are known by (4). Bruno Salvy [12] in a personal communication gave me a method to compute B_n (my personal method fails at $n = 7$).

Denote by $J_n(t) = \int_0^t \sum_{k=0}^{n-1} B_k(u) B_{n-1-k}(t-u) du$, and by TJ_n , TB_n the Laplace transform of J_n and B_n . We have

$$TJ_n(s) = \sum_{k=0}^{n-1} TB_k(s) TB_{n-k-1}(s).$$

It turns out that knowing the first values of B_k , the computation of J_n by the previous formula and by inversion of the Laplace transform ends, using maple (when it does not by simple integration). Then it appears that $J_n(v)$ is a polynomial in $\sin(v)$, $\cos(v)$ and v . The subsequent integration $\int_0^t J_n(v) / \sin(v)^{2n+1} dv$ is possible by helping the computer. Using $\cos(v)^2 + \sin(v)^2 = 1$, it is possible to rewrite J_n as polynomial of degree at most 1 in $\cos(v)$. Then write $J_n(v) / \sin(v)^{2n+1}$ under the

form $Q_0 + Q_1/\sin(v)^1 + \sum_{l=1}^{2n+1} Q_l/\sin(v)^l$, where Q_n is a polynomial in $\cos(v)$ and v . Then, proceed to successive integrations by parts, starting from the largest degree at the denominator till $l = 2$. The remaining integral is computed by a simple integration.

This method allows one to compute 11 terms with maple. The computation of $B_{n,m}$ is possible using the same algorithm, except that some complications arise from the inverse Laplace which makes appear some polylogarithm functions (of the type $\text{polylog}(n, e^{it}) + \text{polylog}(n, -e^{it})$) for some $n \geq 2$). There limit at $t \rightarrow 0$ have to be treated separately, letting $\zeta(n)$ plays a role in the final result. I am able to compute $B_{n,m}$ for $n \leq 5$, and then $\mathbb{P}_{n,m}^D$ for $n \leq 7$.

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